Provenance-Carrying Evidence Fusion for Biomedical Prediction

A typed evidence reducer, and the pre-registration of two experiments

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Abstract

Biomedical prediction increasingly proceeds by fusing heterogeneous evidence: sequence homology, learned embeddings, annotation transfer, and neural link predictors. The quantities being fused are heterogeneous *in kind* — some are ordinal similarities, some are binary observations, few are calibrated probabilities — and they are mutually dependent, because they are computed from overlapping data. Common practice combines them anyway, because they all happen to be numbers in $[0, 1]$.

We describe a platform that refuses to do this. Evidence is typed by kind, and combinations that are not semantically justified simply have no reduction rule: noisy-OR is defined only on calibrated probabilities that share a calibration regime and lie in distinct dependence groups, and evidential revision is defined only on disjoint provenance. The fusion itself is performed by a streaming grouped reduction that emits, for each group, a cryptographic certificate of the evidence it consumed.

This report has two purposes. It documents the platform and the verification it currently passes, and it *pre-registers* two experiments — a partial-knowledge protein function task and an lncRNA–target link prediction task — including their hypotheses, splits, and analysis, before their outcomes are known.

We claim no biological performance in this report. No full evaluation has been run, and no comparison against a trained control exists. Every result table below is generated mechanically from a run receipt; where a run has not been performed, the table says so and reports nothing.

1 Introduction

A prediction that cannot be interrogated is difficult to act on. In biomedical settings the consumer of a prediction usually needs to know not only its score but what supported it, whether the supporting evidence was independent, and how much of the apparent confidence is an artifact of counting the same observation twice. Evidence fusion is exactly where these questions are decided, and it is usually where they are quietly discarded.

Two failure modes recur. The first is *kind confusion*: a BLAST bit-score, a cosine similarity between protein embeddings, and a binary evidence flag are all real numbers in $[0, 1]$, and are therefore combined by formulas — noisy-OR most often — whose derivation assumes they are calibrated probabilities. The formula is well-defined on the floats; it is meaningless on the quantities. The second is *dependence blindness*: predictors that share training annotations or an embedding are treated as independent witnesses, so agreement between them inflates confidence when it should not.

Our position is that both failures are type errors, and should be prevented the way type errors are prevented: by making the illegal combination inexpressible rather than by warning about it in prose.

Contributions

1. A typed evidence discipline (§2) in which ordinal scores, binary observations, calibrated probabilities, and evidential counts are distinct kinds, and unjustified combinations have no reduction rule.
2. A streaming grouped reduction with per-group evidence certificates (§3), in ordered and external-merge modes whose exact agreement is checked by executable differential tests, with bounded source-side input memory under direct cursor ingestion.
3. The pre-registration of two experiments (§4, §5), with hypotheses and analyses fixed in advance.
4. Verification evidence for the platform (§6), including exact agreement of the load-bearing reducer with an independent reference implementation on a real evidence slice.

2 Typed evidence

We distinguish four kinds of evidence. Let g range over *dependence groups* — declared equivalence classes of sources that share data, training annotations, or a model family — and let c range over *calibration regimes*.

Definition 1 (Evidence kinds)**.**

$\mathsf{Ordinal}(s, v)$	*a value v on a named ordinal scale s; comparable, not a probability*
$\mathsf{Binary}(\sigma, o)$	*an observation o recorded by source σ*
$\mathsf{Prob}(c, g, \sigma, p)$	$p \in [0, 1]$ *calibrated under regime c, from source σ in group g*
$\mathsf{PLN}(n^+, n^-, \tau)$	*positive/negative evidence counts with a provenance stamp τ*

The discipline is carried by which combinations are *defined.*

Maximum is scale-local. $\mathsf{Ordinal}(s, v_1)$ and $\mathsf{Ordinal}(s, v_2)$ reduce to $\mathsf{Ordinal}(s, \max(v_1, v_2))$ only when the scale s is literally the same symbol. Two ordinal scores on different scales have no maximum, because their order relations are not commensurable.

Dependence-aware noisy-OR. For $\mathsf{Prob}(c, g_1, \sigma_1, p_1)$ and $\mathsf{Prob}(c, g_2, \sigma_2, p_2)$ the reduction

$$p_1 \oplus p_2 = 1 - (1 - p_1)(1 - p_2) \tag{1}$$

is defined *only if* the calibration regimes agree and $g_1 \neq g_2$. Two consequences matter. An ordinal score can never enter Eq. 1, because it is not of kind Prob and no rule mediates the two. And two sources within one dependence group can never be noisy-OR'd, so declaring sources dependent is a statement with teeth rather than documentation.

Revision requires disjoint provenance. PLN counts revise by addition only when their provenance stamps are disjoint, $\tau_1 \cap \tau_2 = \emptyset$. Overlapping provenance has no revision rule; a policy that resolves the overlap must be selected explicitly by the caller, never guessed.

The engineering content of this section is that "no rule" is the mechanism. There is no runtime flag to override, and no default that silently applies. A pipeline that attempts an unjustified fusion does not produce a wrong number; it produces nothing, and fails.

3 Streaming grouped reduction and certificates

Fusion is a grouped reduction: partition evidence by a key (here, a protein–term pair and its dependence group), reduce each group, then combine groups. At biomedical scale the evidence does not fit in memory, so the reducer must stream its input. Materialized result storage remains a separate consumer-side cost and is not included in the source-memory claim below.

The reducer exposes one surface, `group-fold`, in two modes.

`ordered` requires keys in canonical non-decreasing order. It holds exactly one active accumulator, releases each group when the key advances, and *rejects* key regression rather than silently producing fragmented groups. Ordering is over the canonical text of the key, which is bytewise rather than numeric — a distinction that matters for integer keys and is benign for identifier keys.

`external-merge` accepts arbitrary input order. It builds sorted runs bounded by a configured size, tags every record with a global monotonic ordinal, and performs a stable k-way merge that feeds the same ordered reducer. Equal keys retain encounter order.

Mode-agreement contract. `external-merge` totally orders records by (key, ordinal) and feeds the ordered reducer over the same canonical bytes. The intended contract is that both modes produce identical groups and identical certificates. This is checked by executable differential tests rather than proved formally; the formal obligation is stated in §9.

Certificates. Each emitted group carries $(\text{count}, \text{SHA-256})$ over the canonical byte stream of the items that entered it, with each item length-prefixed before hashing so that the digest is collision-resistant across item boundaries. Together with the retained trail or replayable input, the certificate commits to exactly which serialized evidence produced the number and how many items were consumed; the digest alone does not recover or name those items.

4 Experiment 1: partial-knowledge protein function

4.1 Task and boundary

The task is the partial-knowledge protein function prediction setting: known annotations are frozen at a submission cutoff, and the ground truth consists of annotations newly acquired in a later growth period. The community evaluator is the scoring authority; we do not define the metric.

The temporal boundary is a pre-registered constraint, not an aspiration. The final ground truth may be read *only* by target selection and by the evaluator. It may not fit calibration, dependence groups, thresholds, source weights, or any fusion parameter.

4.2 Status of the released predictors

Publicly released predictors (sequence homology, embedding similarity, annotation transfer) are treated as *frozen input features and controls*, not as our result. Reporting their scores would be a reproduction, not an experiment.

Critically, these released scores are *not* calibrated probabilities: the homology scores are normalised bit-scores, the embedding scores are cosine similarities, and the annotation-transfer evidence is binary. Under the discipline of §2 they are of kind Ordinal and Binary, and therefore *cannot* enter Eq. 1 at all.

4.3 Pre-registered fusion policy

The primary policy is the conservative one: all released sources are placed in a single dependence group and reduced by maximum. This assumes the sources are mutually dependent, which is the defensible default given that they derive from overlapping sequence and annotation data.

A calibrated noisy-OR policy is available but gated: it requires a pinned calibration artifact per source, one shared calibration regime, and an explicit independence declaration for every cross-group pair. Until a calibration is actually fitted from eligible pre-cutoff data, this policy is not reportable, and any result obtained under it is not a result.

5 Experiment 2: lncRNA–target link prediction

5.1 Systems

Three systems are trained on identical frozen splits across five seeds:

1. a freshly trained neural path-reasoning control (no published checkpoint is reused);
2. a relation-path evidence model, which mines typed relation-path templates from the background graph and learns positive/negative evidence per template under a Beta-smoothed count-to-truth semantics;
3. a hybrid, combining the control's logit and the path model's evidence log-odds through a non-negative L_2-regularised logistic stacker.

A naming caution. The relation-path evidence model is *PLN-style* — it uses evidence counts and a count-to-truth reading — but it is presently implemented natively rather than executed in the WM-PLN runtime. We therefore do not call this lane "WM-PLN" and do not claim it as a WM-PLN result. Grounding it in the runtime is stated as outstanding work in §9.

5.2 Leakage controls

Three controls are mechanical rather than documentary.

Self-leakage. When collecting path features for a query edge, the query edge and its inverse are removed from the background graph, so the model cannot observe the answer it is asked for.

Selection isolation. During training and model selection the test split is not merely unread by convention: the selection view substitutes the validation file in place of the test file and records a receipt attesting the substitution. Training and selection commands refuse to run unless that receipt is present and affirms that no real test bytes are staged.

One-use test. The test evaluation is guarded by an exclusive-create lock and a per-job state machine that moves each job from pending to in-progress to consumed. A second claim fails. The lock is claimed before any test byte is opened.

5.3 Pre-registered hypothesis and analysis

Primary hypothesis. The hybrid achieves higher MRR on the held-out test split than the freshly trained neural control.

The primary analysis is a paired per-query bootstrap confidence interval for the MRR difference, with the per-seed distribution reported. Hits@k, mean rank, calibration, the path-model-only lane, provenance coverage, and path ablations are secondary. Selection uses validation only; the test split is scored exactly once. **A non-improvement, or an interval spanning zero, is a valid result and will be reported as such.** No post-test retuning is permitted.

5.4 Negative controls

Negatives are sampled by exact type compatibility, and shortfalls are recorded rather than filled by broadening the type. A sampled negative is a *sampled* negative: it is a type-matched pair not present in the supervision set, not a biologically verified falsehood. Because filtering sampled negatives against held-out edges would itself constitute leakage, a sampled negative may coincide with a true held-out edge. Calibration measured against sampled negatives is therefore calibration against that control distribution, and is not population calibration.

6 Platform verification

The following are engineering results about the platform. They are not biological results.

6.1 Reducer equivalence on a real evidence slice

The load-bearing reducer was run against an independent reference implementation of the same reduction law over a real evidence slice, under the conservative maximum policy, with evidence partitioned into shards by a digest of the protein identifier. Table 1 is generated from the gate's receipt.

Quantity	Value
Primary backend	`cetta-group-fold`
Primary role	`load-bearing-reducer`
Reference backend	`sqlite-differential-oracle`
Fusion policy	`conservative-max`
Input rows	92,202
Evidence packets	92,202
Dependence groups	75,151
Fused candidates	75,151
Exact duplicates collapsed	0
Partition shards (active)	8 (8)
Compared rows	75,151
Maximum absolute error	**0.0**
Prediction digest (primary)	`2235ca4c4a4c...`
Prediction digest (reference)	`2235ca4c4a4c...`
Shard manifest digest	`427d72e55090...`

Evidence source	Rows
all-evidence	7,590
blast-cafa5	74,353
prott5-cosine-5	10,259

Table 1: Differential equivalence of the load-bearing reducer against the independent reference implementation on a real evidence slice. Both implementations consume the same 92,202 input rows and emit byte-identical fused predictions. Status reported by the gate: `passed`.

Two facts are worth drawing out. The maximum absolute error between the two implementations is exactly zero over all compared rows, and the two implementations emit the same prediction digest — so the agreement is byte-level, not merely numerical. The sharded, streaming reducer and a straightforward relational reference compute the same function on real data.

6.2 Pre-registration state

Table 2 evidences that the frozen plan's test policy remains unarmed.

Quantity	Value
Plan fingerprint	`10b2a52df68a...`
Frozen jobs	15
Test policy state	`unarmed`

Table 2: Pre-registration state of the frozen evaluation plan. The test split is reported here solely to evidence that it remains unopened.

7 Preliminary observation, and what is absent

7.1 One validation candidate

Exactly one hyperparameter point of the relation-path evidence model has been trained and scored on validation. It is reported in Table 3 because suppressing a weak preliminary observation would be a selection effect.

Quantity	Value
Job	`pln-path-evidence--seed-1729`
Experiment fingerprint	`10b2a52df68a...`
Model digest	`a85556b90309...`
Path templates	33,656
Depth / min. support / path cap	3 / 2 / 512
Evidence prior (α, β)	(1, 1)
Seed	1729
Supervised positives	4,867
Sampled negatives (requested)	155,744
Sampled negatives (obtained)	155,618
Sampled-negative shortfall	126
Positives without exact-type control	2
Validation queries	608
Candidates per query (mean)	929.2
MRR	0.011627
Mean rank	346.97
Hits@1 / @3 / @10	0.0000 / 0.0000 / 0.0016
Brier vs. sampled negatives	0.043446
Brier examples	20,064
Test split opened	`false`
Test plan state	`unarmed`

Table 3: Single unselected validation candidate of the relation-path evidence model. Reported status: `validation-candidate-complete-not-selected-not-test`. These are validation-only observations for one hyperparameter point; they are not a test result, not a selected model, and not a comparison against any control. Shortfall policy recorded by the run: *use-all-available-and-record; never broaden target type.*

The MRR is weak. Against a mean candidate set of roughly 929 entries, a random ranking would achieve an MRR of approximately $\ln(929)/929 \approx 0.0074$; the observed 0.0116 is therefore on the order of $1.6\times$ random. This is an unselected point in a grid, at one seed, on validation, with no control to compare against. It supports no claim about the method, and we draw none. It is recorded so that the eventual grid cannot be read as if this point had never existed.

7.2 The results that do not yet exist

PENDING: full partial-knowledge evaluation. The frozen full-target CAFA5 evaluation has not been run. No receipt exists for this run, so no value is reported here.

PENDING: control / path-evidence / hybrid comparison. The multi-seed training grid and the one-use test evaluation have not been run, so no comparison against a control exists. No receipt exists for this run, so no value is reported here.

8 What "provenance" does and does not mean here

Precision here is worth more than enthusiasm.

What exists. Every fused prediction carries a trail naming the source packets that contributed, their dependence groups, and their scores; each reduction group carries a recomputable count/digest certificate; the link prediction lane emits witness paths, contributing input-line digests, learned template evidence, and a training-manifest digest. Separately, a formal development of provenance-aware evidential inference exists, covering source and scope support, disjointness, revision unions, and scoped forgetting.

What does not exist. These artifacts do not yet compose into a single verified chain

$$\text{dataset/split} \to \text{training} \to \text{model/calibration} \to \text{inference} \to \text{prediction} \to \text{evaluation receipt},$$

and we do not yet prove that a runtime certificate denotes the corresponding formal derivation. The honest term for the present state is therefore *provenance certificates and audit trails*, not *proven provenance chains*. We use the former throughout, and the latter nowhere.

9 Limitations and threats to validity

Positive-unlabelled ground truth. Protein function annotation is sparse and open-world: the absence of an annotation is not evidence of absence. Precision measured against such a ground truth is a lower bound, and apparent false positives may be undiscovered true positives.

Calibration is not yet fitted. The calibrated fusion policy is plumbed but its calibration has not been learned from eligible data. No calibrated result is reported here, and none may be reported until the calibration is fitted, without consulting final labels.

Scale is designed for, not yet demonstrated. The reducer streams and spills by construction, and has been exercised at 10^5 records and at a real 10^4-scale evidence slice. The full evaluation is roughly three orders of magnitude larger. Working memory is bounded per shard and fails loudly rather than degrading, so a skewed partition is a diagnosable error rather than a silent one — but the full-scale claim is an expectation, not a measurement.

Mode agreement is tested, not proved. The agreement of the ordered and external-merge modes, the once-counting property of the reduction, and the admissibility conditions of Eq. 1 are enforced by executable tests. Discharging them as formal obligations against the evidential semantics is outstanding.

The path model is not yet runtime-grounded. As noted in §5, the relation-path evidence lane is PLN-style but natively implemented; it is not a WM-PLN execution result.

10 Reproducibility

Datasets, the evaluator, and the upstream model are pinned by content digest and revision; a run fails rather than proceeds if a pin does not verify. Target selection, shard partitioning, and negative sampling are seeded and deterministic. Evaluation outputs are hashed semantically — rows and columns canonically ordered before digesting — so that a digest reflects content rather than file-writing order.

Every number in the result tables of this report is emitted by a generator that reads run receipts; the generator is included alongside this document. A table whose receipt is missing renders as an explicit pending notice. This is the same discipline the paper argues for, applied to the paper.

References

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